

Dissertation Proposal: *Designing for Accessibility, Low Literacy and Low Connectivity*

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1. INTRODUCTION

Many people worldwide are excluded from digital public services. Drawing on accessibility research and inclusive design principles, the proposed project designs, builds, and evaluates a redesigned e-government webpage, comparing it against a baseline to generate evidence on what works, for whom, and why.

The remainder of this proposal is structured as follows: Section 2 provides background to the problem; Section 3 sets out the project's aim and objectives; Section 4 provides a critical review of the related literature; Section 5 outlines the methodologies and methods employed; Section 6 analyses risks and ethical considerations; and Section 7 presents the project work plan.

2. BACKGROUND

Low literacy is among the most significant barriers to effective technology use. Globally, an estimated 739 million adults lack basic literacy skills (UNESCO, 2025), and functional digital literacy gaps are considerably broader. For users with limited reading proficiency, text-heavy interfaces create usability barriers. Poor connectivity further degrades the experience.

Non-users of e-government services frequently cite low functional literacy, unfamiliarity with digital bureaucratic language, and distrust of online systems as primary reasons for non-engagement. Older adults, individuals with cognitive disabilities, refugees, recent migrants, and those from lower socioeconomic backgrounds are disproportionately excluded from digital public services, including health, housing, and benefits systems.

Despite the scale of this problem, a significant gap remains in research into accessible, low-literacy-optimised design for public e-government services. This project addresses that gap by designing and evaluating a redesigned interface to generate evidence-based design recommendations applicable across both low-resource and high-resource settings.

3. AIM AND OBJECTIVES

3.1. Aim

The aim of this project is to redesign and evaluate an accessible, low-literacy-optimised public e-government service interface, to provide evidence-based insights into design techniques and accessibility standards that improve task completion for underserved and digitally excluded user groups.

3.2. Objectives and Work Packages

Six objectives will be incorporated to support the research, design, and implementation of the project, as well as the findings, results, and recommendations. The objectives will be converted into major tasks or work packages that have a concrete output ([Table 2](#)). The six objectives include:

- O1.** To conduct a literature review of accessibility and low-literacy design guidelines, identifying a minimum of 5 evidence-based design principles to inform the interface redesign.
- O2.** To design a functional version of an e-government webpage (a redesigned accessible version), implementing at least 4 measurable accessibility improvements such as plain language, iconography, increased contrast, and simplified navigation.
- O3.** To develop a PostgreSQL database schema that captures session, task-event, and outcome data, capable of logging interaction records across simulated user profiles.
- O4.** To conduct a usability evaluation with a minimum of 5-10 participants (and potentially one persona profile of a typical user from the target user group), collecting qualitative and quantitative data on task completion rate, time-on-task, and error frequency across both interface versions.
- O5.** To analyse the collected interaction data using SQL queries and produce at least one visualisation tool (Python/Matplotlib), producing comparative findings between the baseline and accessible interface versions.
- O6.** To critically evaluate the system's effectiveness and limitations, reflecting design decisions against the literature and proposing concrete recommendations for future accessible e-government design.

4. CRITICAL REVIEW OF RELATED LITERATURE

4.1. The Scale and Nature of Low Literacy

“Administrative literacy” among a population is key to positive interactions between e-government services and users. In a digital world, making public administration more accessible by simplifying contact channels with civil servants is crucial (Morte-Nadal and Esteban-Navarro, 2025). People with mild intellectual disabilities or low literacy have distinct cognitive and literacy characteristics, and both can face challenges in comprehending digital information because of low information thresholds, limited reading ability and comprehension, or visual sensitivity (van Calis et al., 2025). The global literacy rate has risen considerably over the past five decades, yet significant gaps persist (Literacy, 2020). In 2024, 739 million adults remain illiterate worldwide, a decline from the previous year, which was 754 million (UNESCO, 2025). While multilingualism is common globally, 40% of the world's population lacks access to education in a language they understand (UNESCO, 2016). With growing immigration and the need to access digital services across borders, many people remain inept to navigate an increasingly digital world.

4.2. Digital Exclusion and E-Government Non-Users

The “digital-by-default” policy trend in government services interaction and engagement can serve many consequences for users who find navigating digital government spaces challenging. Moreover, e-government has been shown to be beneficial to the public by increasing the efficiency and effectiveness of service delivery. Factors influencing user experience of e-government services include, but are not limited to: usability, performance,

privacy, content quality and relevance, personalisation, and effective channels for feedback and support (Alazemi and Al-Shehab, 2024).

The United Nations and European Commission research has ranked the United States and the United Kingdom in the top tier of its e-government readiness index for numerous years. However, despite these achievements, the UK has several barriers to e-government adoption; E-government services are still prone to poor user experience (Carter et al., 2016). Most of the 64 reviewed U.S. federal e-government websites did not meet the Web Content Accessibility Guidelines (WCAG) standards, reflecting a broader global trend of poor compliance across government sites (Aljedaani, 2025). The European Commission has produced various strategies, projects, and policy initiatives to drive digital transformation. According to the Europe's Digital Decade: digital target for 2030, outlines the aim for Europe's digital transition by 2030, ensuring that 80% of citizens acquire at least basic digital skills (Morte-Nadal and Esteban-Navarro, 2025). The e-government development index (EGDI) provides a numerical ranking of e-government development among UN member states.

The US General Accounting Office categorised e-government services into four models: government-to-government, government-to-business, government-to-citizens, and government-to-employees (Aljedaani, 2025). E-government maturity models define best practices for developing and implementing functional services, helping gauge and enhance government capability. Maturity models assess process development across three types: practitioner-developed governmental models; holistic models supporting authorities in project implementation and success projection; and evolutionary models advancing e-government through sequential stages (Aldrees and Gračanin, 2023). Mapping current e-government interfaces against these frameworks identifies design and process gaps that exclude low-literacy, low-connectivity, and accessibility constrained users.

4.3. Language, Culture and Linguistic Diversity as Exclusion Factors

Inclusive design has been identified as a potential means of addressing accessibility challenges during the design process of e-government systems due to the focus on incorporating the “full range of human diversity” (Abdurahman and Kabanda, 2024). Many marginalised individuals, such as seniors, linguistic minorities, rural communities, and those with lower educational backgrounds, struggle to navigate online platforms, complete digital transactions, and interpret e-government services. Within these heterogeneous groups, these users can modify, use, and interact with technology in diverse ways regardless of their cultural background (Gomez-Hernandez et al., 2023). Research suggests that individuals prefer in-person interaction with government personnel, as they perceive this type of engagement as more trustworthy and produce faster results rather than digital services (Djatkiko et al., 2025).

A voice navigation system and real-time language translation are features that can support non-native speakers and users with visual impairments. One notable initiative is Microsoft's AI for Accessibility program, which collaborates with government agencies to develop inclusive solutions for people with disabilities, utilising AI to enhance tools such as speech-to-text, real-time captioning, and adaptive interfaces for those with visual and motor impairments. AI can feel overwhelming, especially for groups who already find it intrusive. However, incorporating AI-powered solutions into e-government services can improve engagement among marginalised groups. For example, Canada's AI assistants help citizens navigate government websites, and the EU's partnership with Microsoft has deployed AI sign

language interpretation tools, improving access for the hearing impaired. These initiatives highlight AI's role in closing accessibility gaps (Djatkiko et al., 2025).

4.4. *Connectivity as a Design Constraint*

Offline architecture is a robust design concept where applications treat network access as an optional feature and reposition network connectivity from a strict requirement to a value-added service (Pothineni, 2024). Progressive Web Apps (PWAs) give public sector organisations, including local councils, a major advantage by allowing users to access services and information even without an internet connection. This ensures consistent service delivery, supports users in areas with poor connectivity, and improves convenience, user experience, and inclusivity.

4.5. *Accessible and Inclusive Design Principles*

Two key frameworks inform accessible web design for older adults. Gomez-Hernandez et al. (2023), synthesising 15 studies, identify simplifying interfaces and enlarging interactive controls as the most critical guidelines, grouping remaining recommendations across five areas: help and training, navigation, visual design, cognitive load, and interaction.

4.6. *Evidence Based Design Principles*

The literature reviewed identifies five evidence-based design principles with direct implications for accessible interface redesign. First, **simplification and cognitive load reduction** are supported by both Morte-Nadal and Esteban-Navarro (2025) and Gomez-Hernandez et al. (2023), whose synthesis of 15 empirical studies identifies simplifying content and enlarging interactive controls as the two highest-priority “Golden Rules” for users with low literacy or cognitive impairment. Second, **universal and inclusive design**, defined by Abdurahman and Kabanda (2024) as incorporating the full range of human diversity into the design process, provides a structural principle for addressing the heterogeneous needs of marginalised users, including older adults, linguistic minorities, and those with limited digital literacy. Third, **multimodal and multilingual support** emerges as a distinct principle, with Djatkiko et al. (2025) identifying voice navigation and real-time language translation as evidence-based features that reduce exclusion for non-native speakers and users with visual or motor impairments. Fourth, **offline-first or connectivity-resilient architecture** is identified by Pothineni (2024) as a positive design response to the real-world constraint of limited internet access, demonstrating that Progressive Web App frameworks can enhance usability and service resilience by enabling consistent access for users in low-connectivity environments. Fifth, adherence to the **Web Content Accessibility Guidelines (WCAG) POUR principles**, Perceivable, Operable, Understandable, and Robust, provides a standardised, internationally recognised framework within which the preceding principles can be operationalised (Aljedaani, 2025). These five principles form an evidence-based foundation for the interface redesign proposed in this study.

5. METHODOLOGIES AND METHODS

5.1. *Introduction*

The proposal evaluates a redesigned, accessibility-focused webpage by comparing it with a baseline version and analysing its effectiveness across different user groups, including low-literacy and digitally excluded users. To achieve this, the project employs a

mixed-methods and design science research (DSR) approach, combining creation and evaluation of a prototype webpage and quantitative interaction logging with qualitative usability research and testing techniques. This approach acknowledges the building of the webpage as a research contribution, not just the testing and follows best practice in accessibility and e-government research, as highlighted in the literature review on inclusive digital service design (Islam et al., 2023; van Calis et al., 2025; Abdurahman & Kabanda, 2024).

5.2. Research Design Rationale

The Design Science Research (DSR) methodology (Hevner et al., 2004; Peffers et al., 2007), frames research around the design, construction, and evaluation that addresses a defined real-world problem. DSR is suited to applied computing and HCI research because it treats the built system (the accessible e-government prototype and PostgreSQL interaction) as a primary research contribution, and not as preparation for the study. A mixed-methods design is also considered to capture both behavioural performance data and user-perceived barriers. Quantitative measures, such as task completion rate, time-on-task, and error frequency, enable objective comparison of usability outcomes between the baseline and redesigned versions. Qualitative insights gathered through think-aloud protocols, interviews, and observational notes provide a deeper understanding of users' mental models, comprehension challenges, and trust perceptions. This approach aligns with findings that e-government adoption is shaped by both measurable usability barriers and subjective experiences such as trust, cognitive load, and perceived complexity (Carter et al., 2016; Ueno et al., 2023).

5.3. Prototype Development

The prototype functions as a testbed to evaluate accessibility enhancements and the impact of design changes under conditions that reflect real-world constraints. For the website front-end, HTML, CSS, and JavaScript will be used in the design. To identify a suitable webpage for redesign, preliminary research will include screening public-facing e-government webpages in the U.S. at the county level, focusing on low-income and rural counties across the United States. Candidate webpages will be evaluated against WCAG 2.2 accessibility criteria using automated checkers and visual inspection to assess suitability for the project scope. The prototype will consist of a selected county-level service page redesigned implementing evidence-based improvement. A PostgreSQL database will log user interactions with both versions to support the quantitative part of the study.

Guided by accessibility standards (WCAG 2.2) and low-literacy design principles (Islam et al., 2023; van Calis et al., 2025), the redesigned interface will include: large, high-contrast text, icon-supported navigation, plain-language explanations, step-by-step guided tasks, Progressive Web App (PWA) offline capability, reduced form complexity and fewer input fields, visual progress indicators, and support for multiple language variants or simplified English. Offline-first development approaches recommended in recent literature (Pothineni,

2024; Josephe et al., 2023) are incorporated to ensure usability under low connectivity or rural areas.

A PostgreSQL database will be used behind the prototype to capture user interactions. This logging enables detailed behavioural analysis and supports the quantitative dimension of the study. Its primary purpose is to record how long users take to locate information, as well as to log accessibility-relevant behavioural markers while navigating the website. For the website back-end and prototype interaction, the server that connects the website to PostgreSQL, Python will be used. Other frameworks, such as Flask or Django, will be considered.

The logging system tracks time-on-task, navigation patterns (clicks, errors, retries), accessibility markers, and enables comparison across user groups and interface versions. Session data, demographics, event logs, and task outcomes, including completion status, error counts, and abandonment rates, will also be captured.

5.4. Target Population

Participants will initially be selected to reflect groups that commonly face digital exclusion: Adults with low literacy or semi-literacy, older adults, migrants and users with English as a second language, individuals with low digital confidence, and citizens who rarely use online government services. Individuals will be excluded if they are IT professionals, have expertise in UX or usability research, or are already familiar with the content or structure of the prototype, as such prior knowledge could introduce bias into the evaluation. A sample of 5-10 participants is considered adequate for usability-focused mixed-methods studies, where depth of insight outweighs statistical generalisability (Nielsen, 1994).

5.5. Data Collection Methods

Qualitative methods for data collection include brief interviews to investigate clarity of language and icons, trust and confidence in the system, previous experiences with e-government, connectivity frustrations, and preferred features. Such qualitative investigations reflect best practice in studying marginalised user groups (Djatkiko et al., 2025; Allmann & Radu, 2023). Additionally, the “Think-Aloud Protocol” where participants verbalise what they understand, what confuses them, and why they click certain elements will be implemented. This is particularly important for low-literacy users, as comprehension challenges often occur internally (Islam et al., 2023).

Using Python and SQL queries, data will be analysed as follows: Descriptive statistics, to measure mean task time, error count, completion rate; and demographic comparisons of literacy level, age, and language influence.

5.6. Experimental Procedure

Step 1: Introduction and Consent (participants are briefed using a plain-language, icon-supported information sheet, following ethical recommendations for low-literacy participants); **Step 2:** Pre-Study Questionnaire (collects demographic and background data); **Step 3:** Task Execution on Baseline Version (participants complete tasks while think-aloud

data and behaviour logs are captured); **Step 4:** Task Execution on Accessible Version (same tasks repeated under the same connectivity conditions); **Step 5:** Interview (participants reflect both versions). **Step 6:** Debriefing (participants are thanked and provided with contact information).

6. ANALYSIS OF RISKS AND ANY ETHICAL ISSUES

6.1. Risk register

[Table 1](#) identifies the principal risks in the project's progress, assesses the likelihood and potential impact, and outlines the steps required to obtain ethical approval before participant testing begins.

Table 1

Risk	Likelihood	Impact	Mitigation
Difficulty recruiting target participants	High	High	Recruit via project advisor; widen inclusion criteria to any adult with low digital confidence
PostgreSQL infrastructure issues	Medium	Medium	Build and test from the start; use version control
Scope creep	High	High	Work with advisor; acknowledge any limitations and modifications
Small sample size limits statistical generalisability	High	High	Acknowledge limitations; lean on qualitative data
Connectivity simulation not replicating real-world conditions	Medium	Low	Document gaps between simulated and real conditions
Testing bias	Medium	High	Use standardised task scripts; acknowledge observer effect as a limitation
Delays due to ethical approval issues and personal commitments	Medium	High	Submit ethical application early in the project timeline
Acclimatisation to new tech	Medium	Medium	Utilise additional online tools; seek help from advisor

6.2. Ethical approval

An application will be submitted to the University of East Anglia's Ethics Monitor system, detailing the nature of the study, the participant population, the data collection methods, and the measures in place to protect participant wellbeing and confidentiality. All research will be conducted to follow the conditions set out in the approved ethics application and in line with the University of East Anglia's ethical guidelines for research involving human participants.

7. PROJECT WORK PLAN AND ACCOMPANYING GANTT CHART

The project runs from 26 January to 20 August 2026, across six work packages: WP1 (Proposal and Literature Review), WP2 (Interface Design, HTML/CSS prototypes), WP3 (Database Build, PostgreSQL schema and test data), WP4 (Usability Testing, protocol and dataset), WP5 (Analysis and Visualisation, SQL queries and charts), and WP6 (Evaluation and Write-up, discussion chapter and recommendations). An Ethics Application runs in parallel, with approval expected by 9 June 2026. These phases move the project from initial scoping to a tested, evaluated system, and a set of actionable recommendations for accessible e-government design.

Table 2

Work Packages			
WP	Title	Linked Objective	Key Deliverable
WP1	Literature Review and Design Principles Proposal	O1	Written review; shortlist of design principles
WP2	Interface Design and Prototyping	O2	One functional HTML/CSS web prototype
WP3	Database Design and Implementation	O3	PostgreSQL schema; populated database with interaction logs
WP4	Usability Testing	O4	Testing protocol; raw dataset of session/task/outcome
WP5	Data Analysis and Visualisation	O5	SQL query; dashboards comparing both interface versions
WP6	Evaluation and Critical Reflection	O6	Discussion; design recommendations

CMP-7027X Dissertation - Project Gantt Chart
 26 January 2026 - 20 August 2026 | Proposal due: 30 April | Dissertation due: 20 August

Task / Work Package	Start date	End date	Duration (days)	Key deliverable
				26 Jan 2 Feb 9 Feb 16 Feb 23 Feb 2 Mar 9 Mar 16 Mar 23 Mar 30 Mar 6 Apr 13 Apr 20 Apr 27 Apr 4 May 11 May 18 May 25 May 1 Jun 8 Jun 15 Jun 22 Jun 29 Jun 6 Jul 13 Jul 20 Jul 27 Jul 3 Aug 10 Aug 17 Aug

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